

Capital Gains Dynamics: Model and Results

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Summary

- We replaced Tax-Simulator’s top-down capital-gains realization elasticity with a ground-up behavioral model. Taxpayers now *choose* when to sell out of an explicitly modeled stock of unrealized gains. The decision depends on tax rates, the tax treatment of gains at death, mortality risk, and time preference. Baseline and scenario realization paths are both outputs.
- This buys us three things the old approach couldn’t do:
 - **Intertemporal consistency.** Realizations today reduce next year’s stock of unrealized gains. The long-run accrual-realization contradiction Natasha and Jane have flagged independently about the elasticity approach goes away.
 - **Tax treatment at death as an explicit policy lever.** Step-up, carryover, and deemed realization enter the holder’s deferral payoff directly. We can score regime changes without manually adjusting elasticities.
 - **Heterogeneous responsiveness.** Older, gain-rich taxpayers respond more strongly than younger ones. Mortality dominates their deferral payoff.
- **Two empirical targets pin down the model.** A long-run permanent semi-elasticity of -2.52 and a short-run announcement-year response of twice that. They identify two parameters: how responsive the marginal deferral decision is, and how much of baseline realizations can retime in response to anticipated rate changes. Everything else is fixed by external data or assumption.
- **Headline ten-year scores, 2026–2035 (\$B):**
 - \$156B for a +2pp rate hike on LTCG / qualified dividends, vs. \$115B at JCT.
 - \$240B for carryover basis at death across all five tracked asset classes, vs. \$219B JCT.
 - \$956B for deemed realization at death across all five tracked asset classes, vs. \$595B JCT.

Table 1 has the full set.

1 Background and motivation

When deciding whether to sell appreciated assets, taxpayers weigh nontax benefits — portfolio rebalancing, liquidity for consumption, etc. — against the tax costs. They can sell today, defer to a later date and pay less tax in PV, or hold to death and pay nothing at all if basis is stepped up. The choice depends on unobserved preferences, the stock of accumulated unrealized gains, and expectations about future income, mortality, and tax policy.

When tax rates change, the tax price of selling changes and realizations respond. That responsiveness is the *realization elasticity*, typically estimated as a single empirical parameter.

Tax-Simulator currently scores capital gains policy changes by computing record-level MTR changes, applying a uniform elasticity to get a percent change in each record’s realizations, and recalculating taxes on the adjusted realizations.

This “top-down” or “reduced-form” method is common across the scorekeeping community. It’s downstream of data and computational constraints: in the IRS PUF we lack a longitudinal view of taxpayers and have no information on the stock of unrealized gains.

It’s probably good enough for approximate ten-year cost estimates of straight rate changes. But it has several meaningful shortcomings:

- **Intertemporal accrual-realization dynamics are crudely modeled and possibly contradictory.** For decades Jane Gravelle has argued that the permanent elasticities used by scorekeepers imply realizations exceeding accruals in the long run, which is obviously impossible. Sarin, Summers, Zwick & Zidar (2022) make a related argument: “permanent” elasticities are estimated on time horizons too short to observe deferred realizations. The top-down approach can’t engage with these questions because it takes no view on where new realizations come from or where forgone ones go.
- **It does not admit changes to tax treatment at death.** The deferral decision should depend on how gains are taxed at death. But a single elasticity estimated on current-law behavior excludes those dynamics by construction. We have to manually adjust the elasticity to score carryover or deemed-realization counterfactuals, and can’t model the mechanical revenue effects of those regimes at all.
- **Responsiveness is homogeneous.** A single elasticity applies to every taxpayer regardless of income or age. Responsiveness should depend on those features — especially under regime changes at death, where the incentive to defer varies with mortality risk.

The following sections lay out a ground-up realization model for Tax-Simulator that solves these problems by imputing unrealized gains and mortality rates for all records; explicitly modeling the sell-or-defer decision in terms of tax rates, treatment at death, and mortality; tracking realizations through an age-cohort pseudo-panel so policy-induced realizations this year show up in next year’s gain stock; and calibrating to aggregate elasticity and timing moments.

2 Model

The model has four moving parts: a per-dollar Bellman that picks a representative age-cell agent’s realization rate, a three-bucket split that separates that choice from involuntary and retiming-driven realizations, a bathtub recurrence that tracks the stock of unrealized gains, and two calibration targets that pin down the free parameters. This section describes the model at a high level, leaving the full details to the appendix (D, E).

2.1 Per-dollar Bellman

The dynamic unit is an age-year cell (a, t) — a synthetic cohort constructed from the repeated cross-section, since Tax-Simulator doesn’t track individual taxpayers over time. For each environment $j \in \{B, S\}$ (baseline / scenario), the cell solves the per-dollar problem

$$W_{a,t}^j = \max_{r^D \in [0, 1 - r_{\text{exog}}]} \left\{ \kappa_{a,t} r^D - \frac{\psi}{2} (r^D)^2 - \tau_{a,t}^j r^D + (1 - r_{\text{exog}} - r^D) \beta \left[(1 - m_{a,t}) W_{a+1,t+1}^j + m_{a,t} F_{a,t}^j \right] \right\}, \quad (1)$$

with interior FOC

$$r_{a,t}^{D,j} = \frac{\kappa_{a,t} - MC_{a,t}^j}{\psi}, \quad MC_{a,t}^j = \tau_{a,t}^j + \beta(1 - m_{a,t}) W_{a+1,t+1}^j + \beta m_{a,t} F_{a,t}^j, \quad (2)$$

clipped to $[0, 1 - r_{\text{exog}}]$. The pieces: $\kappa_{a,t}$ is a cell-specific intercept recovered from baseline so that $r^{D,B}$ matches the observed ordinary-bucket realization rate (it’s a calibration nuisance parameter, not an economic primitive); ψ is a single global curvature parameter calibrated against an aggregate elasticity moment; $F^j = (1 - c_\phi^j) \tau^j$ is the death-state tax-liability forgiveness value, where c_ϕ^j is the share of the cell’s gain stock the holder “loses” (in terms of forgiveness) under the prevailing death regime; $m_{a,t}$ is mortality weighted by gain dollars (not by taxpayer count, because the Bellman is per-dollar); and β is a year-varying real discount factor built by Fisher-deflating the 10-year Treasury yield by year- t YoY CPI-U. The Bellman is solved by backward induction on an extended age grid [18, 119], with an SSA Trustees life-table tail past age 80. Full specification in Appendix B.

2.2 Three realization buckets

Aggregate realizations partition into

$$r_{a,t} = \underbrace{\phi_I r_{a,t}^B}_{\text{fixed}} + \underbrace{r_{a,t}^{\text{planned}}}_{\text{retimed}} + \underbrace{r_{a,t}^D}_{\text{Bellman}}. \quad (3)$$

The fixed share captures sales driven by liquidity shocks, business exits, divorce, household moves, and other tax-inelastic events. The planned bucket is a pure conservation-of-dollars retiming channel: a ϕ_P share of baseline dollars can move within a one-year window in proportion to the year-over-year tax-rate differential, normalized by a 5pp reference wedge. The Bellman handles permanent realization responses on the residual. The fixed and planned shares together set the upper bound on the Bellman’s choice set, $r_{\text{exog},B} = (\phi_I + \phi_P) r_{a,t}^B$. The planned bucket is what lets the model produce announcement-year spikes (and clean pull-back/payback under temporary shocks) without disturbing the permanent response — and it’s what the +5.04 short-run target identifies. See Appendix C.

2.3 Bathtub recurrence

A synthetic-cohort law of motion tracks the policy-induced deviation $\Delta G_{a,t} = G_{a,t}^S - G_{a,t}^B$ from the baseline gain stock:

$$\Delta G_{h,t+1} = \sum_a A_{a \rightarrow h} (1 - m_{a,t}^{\text{eff}}) \left[(1 - r_{a,t}^S) \Delta G_{a,t} + G_{a,t}^B (r_{a,t}^B - r_{a,t}^S) \right] + \delta_{\text{route}} \sum_a \omega_{a \rightarrow h} m_{a,t}^{\text{eff}} (G_{a,t}^B + \Delta G_{a,t}). \quad (4)$$

The first bracket is the survivor channel (lock-in / unlock); the second is the heir-routing channel, active only under carryover, with ω the SCF-2022 dollar-weighted heir-age distribution. An exogenous gain-creation inflow common to baseline and scenario cancels in ΔG , which is why the bathtub can track only the deviation and doesn’t need to compute that inflow explicitly. See Appendix E.

2.4 Calibration targets

Two empirical moments identify two free parameters via nested bisection (Appendix F):

- **Long-run** permanent semi-elasticity $d \log R / d\tau = -0.62 / 0.238 \approx -2.52$, measured at simulation year 30 of a +1pp permanent shock. Identifies ψ .
- **Short-run** announcement-year response $d \log R(t) / d\tau(t+1) = +5.04$, measured at the announcement year of a delayed +5pp shock. Identifies ϕ_P .

The long-run target is the literature arc elasticity (-0.62) anchored at a fixed baseline $\tau = 0.238$, matching the convention used by the Old TBL method. The short-run target is twice the long-run magnitude with sign flipped (future-tax-up $\Rightarrow$ realize-today), which is roughly what’s needed to match empirical announcement-year asymmetry. The cell-level intercepts $\{\kappa_{a,t}\}$ are recovered from a backward FOC inversion at the start of each baseline/scenario pre-pass. Production values: $\psi = 26.3535$, $\phi_P = 0.3041$.

3 Results

3.1 Scores: six reforms, 10-year window 2026–2035

Table 1 reports 10-year revenue under New TBL method for the six reforms we’re running through the new model: a +2pp top-rate increase on LTCG / qualified dividends, carryover basis at death, deemed realization at death, the two rate-plus-regime combinations, and a +2pp change announced in 2026 and effective in 2027. Each row also reports the Old TBL method’s score and the published JCT estimate where one exists.

Table 1: 10-year revenue impact, \$B (2026–2035).

Reform	New TBL method	Old TBL method	JCT
+2pp on LTCG / qualified dividends (step-up baseline)	156	143	115
+2pp on LTCG (marginal, on carryover baseline)	199	–	–
+2pp on LTCG (marginal, on deemed baseline)	284	–	–
Carryover basis, all 5 classes	240	–	219
Deemed realization, all 5 classes	956	–	595
+2pp announced 2026, effective 2027	130	129	–

A few items to note:

- The rate hike score is close to that of our old approach by construction: we calibrate to the same long-run elasticity (about 0.62 in full-elasticity terms). We’re slightly higher than JCT, who uses a higher elasticity (I believe 0.7).
- Our carryover score is close to that of JCT.
- Our deemed realization score is much larger than JCT’s, a factor of about 1.6x. Somewhat puzzling to me. My best guess is that they are building in more avoidance or evasion, likely based on the estate tax, where “what shows up on the 706” about half that of “what shows up on SCF multiplied through mortality rates” – a discrepancy which I’ve always interpreted as end-of-life planning, trusts, the IRS’s inability to value certain assets, etc. We make no such adjustment beyond the charitable bequest avoidance margin. If we applied a similar

adjustment to gains we'd be close to their number. I need to think about this piece. I think we should circulate to them and see what they think.

- The marginal revenue impact of raising rates against a baseline of carryover or deemed realization is larger than against current law. The score rises from \$156B under step-up to \$199B under carryover and \$284B under deemed realization. This is a huge feature of the model (quantified below in terms of elasticities) – without step-up, you can raise more revenue by raising rates because the main avoidance margin is gone.

The +2pp rate hike comes in at \$156B, 9% above the Old TBL method and 36% above JCT. The gap to the Old TBL method is the New TBL method's behavioral feedback through the bathtub — the stock of unrealized gains responds to deferral, which compounds over time. The JCT gap is harder to pin down without their workpapers, but the New TBL method produces a stronger realization response per pp than the JCT scorekeeping convention does.

Carryover (\$240B) lands close to JCT (\$219B), within $\sim 10\%$. This is the cleanest comparison in the table: a regime change with no rate component, scored by both models, with similar order of magnitude.

Deemed (\$956B) is well above JCT (\$595B), by about 60%. The new terminal-charity model trims taxable death stock by roughly the share of decedent gain dollars routed to charity, but doesn't fully close the gap to JCT. The remainder reflects the model's older, gain-rich cohorts where mortality is high and the loss of step-up forgiveness bites immediately — the same channel that produces the strong $|\text{elasticity}|$ ranking by regime in §3.2. Pulling deemed down further would likely require either an explicit $c_\phi^{\text{deemed}} < 1$ (to reflect surviving-spouse step-up retention, family valuation gaming, etc.) or per-asset ψ heterogeneity that weakens the response on housing.

The delayed announcement comes in at \$130B under the New TBL method vs. \$129B under the Old TBL method — effectively the same as the permanent +2pp under either model. The announcement-year realization spike (planned-bucket dollars pulled forward from 2027) and the 2027 payback offset within the 10-year window. The Old TBL method produces the same outcome because it has no intertemporal mechanism that retimes dollars; the New TBL method produces the same total despite *having* that mechanism, which is the correct answer.

3.2 Implied long-run elasticity, by death regime

A direct consequence of the structural setup is that the model's implied long-run realization elasticity isn't a fixed parameter. It varies with the death regime.

Step-up gives the largest reward for deferring: hold to death and the liability is forgiven outright. A rate hike widens the gap between selling now and deferring to a tax-free exit, so deferral becomes much more attractive. Deemed realization gives no reward — death is just another forced sale — so the response is muted. Carryover sits in between.

Table 2 reports the implied semi-elasticity at sim year 30 (the long-run steady state, where the bathtub has fully rolled forward) of a +2pp permanent shock to LTCG / qualified-dividend rates. The full-elasticity column converts via the literature anchor $\tau = 0.238$.

Table 2: Sim-year-30 implied realization elasticity, by death regime, under a +2pp permanent shock to LTCG / qualified dividends. Full elasticity is the semi-elasticity multiplied by the literature anchor $\tau = 0.238$.

Death regime	Semi- ($d \log R / d \tau$)	Full ($d \log R / d \log \tau$)
Step-up (current law)	-2.5	-0.60
Carryover	-1.8	-0.43
Deemed realization	-1.0	-0.23

Step-up lands at -2.5 (full elasticity -0.60), matching the calibration anchor by construction. The drop to -1.0 (-0.23 full) under deemed is the new result — it falls out of the model’s structure, not from any direct calibration. The size of the rate response depends on the size of the deferral payoff, and that payoff is mechanically smaller under regimes that tax at death. Under step-up the holder pockets the full τ by holding to death, so a rate hike directly widens the sell-now-vs-defer wedge. Under deemed, holding to death pays nothing extra relative to selling now, so the rate hike’s effect on the deferral decision is muted.

This is a useful diagnostic for elasticity-based scorekeeping. The literature’s -0.62 arc elasticity is estimated on a world where step-up is the prevailing regime. Applying the same elasticity to score a rate change *combined with* carryover or deemed would overstate the realization response — by roughly 30% under carryover and more than half under deemed — and overstate the revenue.

3.3 Channel decomposition

Figure 1 shows the year-by-year revenue decomposition for each reform. The engine’s conventional ΔR is attributed to four channels, year by year:

- **Mechanical rate.** $(\tau^S - \tau^B) R^B$: baseline realizations taxed at the new rate.
- **Realization-rate response.** $\tau^S G^B(r^S - r^B)$: lock-in and unlock from both the Bellman and planned-timing buckets, applied to the baseline gain stock.
- **Revenue on policy-induced stock.** $\tau^S r^S \Delta G$: nonzero only under death-regime changes, where the bathtub redistributes or augments the stock relative to baseline.
- **Deemed at death.** The decedent-weighted tax base under deemed regimes, net of the §121 carve-out and the terminal charity adjustment.

The residual between the engine ΔR and the sum of the four channels is reported as a nonlinearity / bracket-interaction wedge.

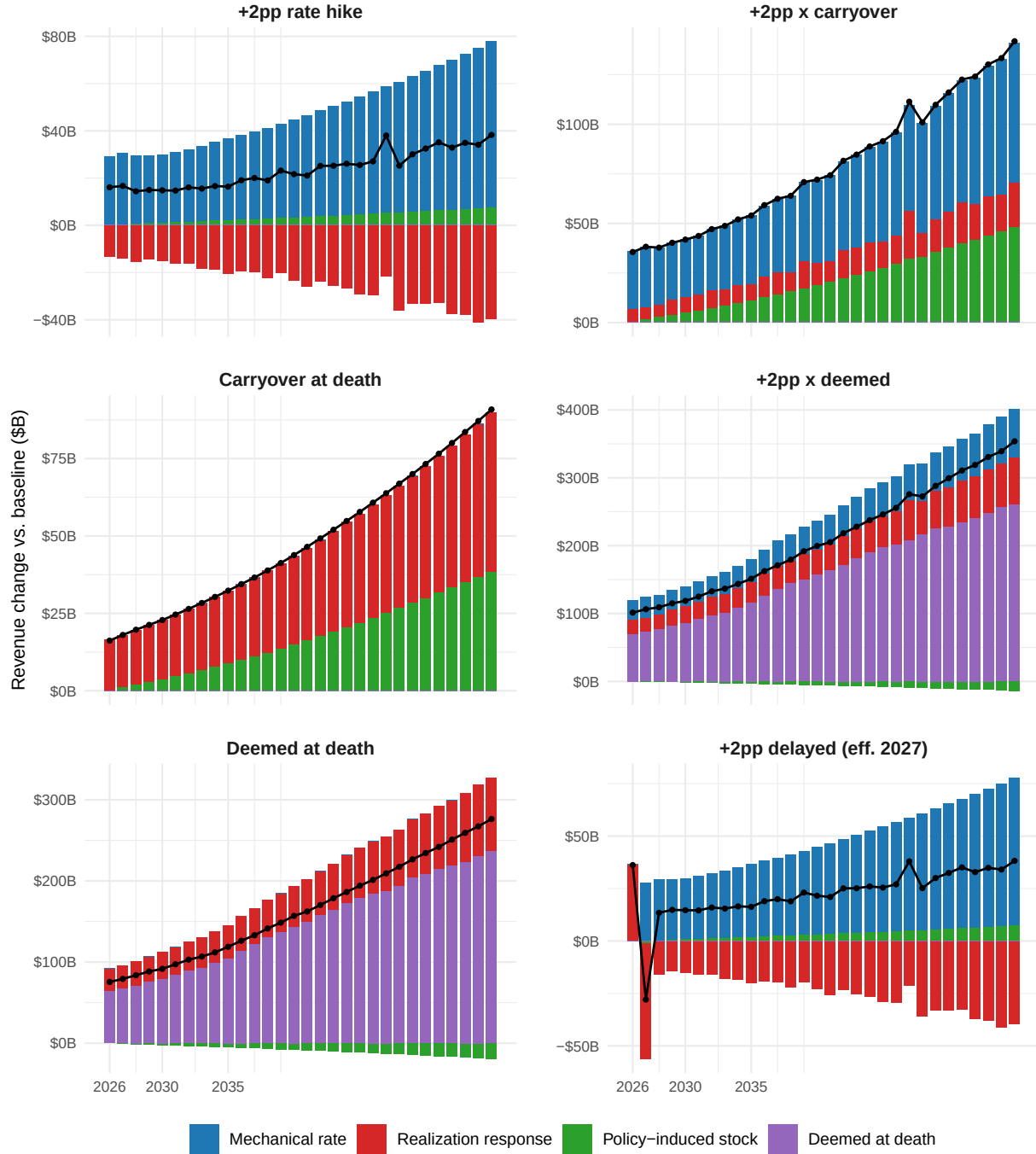


Figure 1: Year-by-year revenue decomposition for the six reforms. Stacked bars are the cell-level channel attribution; the black line is the engine's conventional ΔR from the microdata pass. The bars sum to within $\sim 1\%$ of the engine line; the small gap is from bracket / NIIT / AMT non-linearities the cell algebra can't see.

A Notation

a, t	age cell, simulation year
$j \in \{B, S\}$	baseline / scenario environment
$G_{a,t}^j$	unrealized-gain dollar stock
$R_{a,t}^j$	realized positive long-term gains
$r_{a,t}^j = R^j / G^j$	total cell realization rate
$r_{a,t}^D$	ordinary (Bellman-controlled) discretionary rate
ϕ_I, ϕ_P	fixed share, planned share of realizations
$\tau_{a,t}^j$	average long-term-gains marginal tax rate
$m_{a,t}$	gain-weighted cell mortality
β_t	year-varying real-rate discount factor
$W_{a,t}^j$	per-dollar Bellman value
$\kappa_{a,t}$	per-dollar nontax-benefit intercept
ψ	global curvature parameter
$c_\phi^j(a, t)$	holder-internalized death-stock burden share
θ	heir-burden internalization weight (carryover)
$F_{a,t}^j = (1 - c_\phi^j)\tau^j$	death-state forgiveness value
$\Delta G_{a,t}$	policy-induced gain-stock divergence
$\omega_{a \rightarrow h}$	SCF-derived dollar-weighted heir-age distribution
$A_{a \rightarrow h}$	age-aging operator (one-year increment)

B Bellman

B.1 Modeling closure

Tax-Simulator is a repeated cross-section, so New TBL method can't maintain record-level dynamic wealth histories. The natural dynamic unit is the representative age-year cell, treated as a synthetic cohort. The Bellman is normalized to one dollar of unrealized gain in cell (a, t) ; the bathtub then scales chosen realization rates by the cell's actual dollar stock. Baseline gain-stock growth is treated as an additive exogenous inflow that cancels in the ΔG recurrence between scenario and baseline (Appendix E). The model is partial equilibrium: tax payments are absorbed by an unmodeled outside margin (lower consumption, lower other saving, borrowing, or higher labor supply); the discount factor β prices that outside-margin timing cost. Asset allocation, portfolio returns, and dynastic heir choices are out of scope.

B.2 Primitives

For each cell the baseline pipeline supplies $G_{a,t}^B, R_{a,t}^B, r_{a,t}^B, m_{a,t}, \tau_{a,t}^B$; scenario rates $\tau_{a,t}^S$ and a per-asset death-regime vector come from the policy YAML. The discount factor series $\{\beta_t\}$ is constructed by Fisher-deflating the ten-year-Treasury yield `tsy_10y` by year- t YoY CPI-U from Macro-Projections (the user memo `feedback_kg_dynamics_real_rates` documents the decision to use a real discount factor). Mortality $m_{a,t}$ is gain-stock-weighted at the cell level: $m_{a,t} = \sum_i w_i m_i^{\text{hh}} G_i^{\text{unit}} / \sum_i w_i G_i^{\text{unit}}$, where m_i^{hh} is per-household ($q_1^{\text{death}} q_2^{\text{death}}$ for joint filers, q_1^{death} otherwise). This corrects a known bias in earlier specs that taxpayer-weighted mortality overstated m by 2–3 $\times$ relative to the per-dollar mortality the Bellman should use.

The age grid is extended past the bathtub topcode (age 80) to 119 using an SSA Trustees life table (Alternative 2, 50/50 male/female blend), with $r^B(a) = r^B(80)$ held flat in the tail so the continuation value of an 80-year-old is anchored to the empirical topcode-pool realization rate rather

than the model-internal zero.

B.3 Equation and FOC

The Bellman (1) prices, per dollar of unrealized gain in (a, t) :

- the nontax benefit of realizing now, $b(r^D; \kappa, \psi) = \kappa r^D - (\psi/2)(r^D)^2$;
- the contemporaneous tax cost $-\tau^j r^D$;
- the discounted continuation value of dollars that survive and remain unrealized, $(1 - r_{\text{exog}} - r^D) \beta (1 - m) W_{a+1, t+1}^j$;
- the discounted death-state forgiveness value, $(1 - r_{\text{exog}} - r^D) \beta m F^j$.

The quadratic benefit is not mandatory but is the transparent first prototype: κ shifts the level of baseline realization motives, ψ controls response curvature, and the interior FOC (2) is closed-form. At corner cells with $r^{D,B} = 0$ the inversion sets $\kappa = MC^B$ exactly so the cell sits at the lower corner under baseline.

B.4 Why the prior log-utility Bellman was discarded

An earlier version of this model used a per-dollar log-benefit Bellman with an exponential response rule $r^{D,S} = r^{D,B} \exp\{-\eta(P^S - P^B)\}$, where P^j is a tax-price wedge constructed from W^j . The trouble is that once the cell intercept κ is recovered from the baseline FOC, the implied response is insensitive to η — it cancels out of r^S/r^B . The quadratic-direct-control formulation in (1) sidesteps this by keeping the level parameter (κ) and the curvature parameter (ψ) algebraically separate.

C Three-bucket realization timing

C.1 Bucket shares

Total cell realizations are partitioned by (3). The fixed share $\phi_I = 0.40$ is held outside the Bellman's choice set and outside the planned-timing channel. The planned share ϕ_P is calibrated jointly with ψ against the short-run target (Appendix F); its production value is 0.3041. The ordinary Bellman bucket therefore controls $(1 - \phi_I - \phi_P) r^B \approx 0.296 r^B$ of baseline realizations. The upper bound on the Bellman control is $r_{\text{exog},B} = (\phi_I + \phi_P) r^B$.

C.2 Planned-timing mechanics

For each cell and year t , a fraction $\xi_{t,t'} = \text{clip}\left(\frac{\tau_t^S - \tau_{t'}^S}{w_{\text{ref}}}, 0, 1\right)$ of the planned-bucket dollars in year t is shifted to a nearby year $t' \in \{t-1, t, t+1\}$ (timing window = 1), where $w_{\text{ref}} = 0.05$ is the reference wedge: a 5pp tax differential moves the full bucket; a 1pp differential moves 20%. Dollars are conserved: shifts sum to zero across the window. The planned bucket has no permanent effect on aggregate realizations and produces clean pull-back/payback patterns under temporary shocks and announcement-effective splits.

C.3 Identification

The two calibration moments are orthogonal in the limit of a small permanent-only shock: ϕ_P is silent on the long-run response (planned dollars retime within a window but do not change the long-run flow) and ψ is silent on the announcement-year response in the limit of zero ϕ_P . Nested bisection (outer: ϕ_P ; inner: ψ) converges in ~ 10 – 20 outer $\times$ 5–10 inner iterations.

D Death channels

D.1 Forgiveness value F and the regime triplet

At death, the holder values the cell at $F^j = (1 - c_\phi^j) \tau^j$, where $c_\phi^j \in [0, 1]$ is the burden share the current holder internalizes on dollars whose realization is being decided today. Three canonical regimes:

$$F = \begin{cases} \tau, & \text{step-up} \quad (c_\phi = 0) \\ (1 - \theta)\tau, & \text{carryover} \quad (c_\phi = \theta) \\ 0, & \text{deemed} \quad (c_\phi = 1). \end{cases} \quad (5)$$

Under step-up, death extinguishes liability for the current holder, so holding gains into death is worth the full τ . Under deemed realization, no liability is forgiven and the death option has zero value. Carryover sits between: liability isn't extinguished at the family level, but the current holder may not fully internalize the heir's future tax burden. The parameter $\theta \in [0, 1]$ governs that internalization; $\theta = 1$ makes carryover look like deemed to the holder; $\theta = 0$ makes it look like step-up. Production default is $\theta = 0.5$, reported as a sensitivity since the empirical literature doesn't pin it down sharply. Note that F is the *holder's* reduced-form valuation; physical stock routing is a separate object that lives in the bathtub (Appendix E).

D.2 Per-asset regime triplet

Five wealth classes are tracked separately: equities, pass-throughs, primary home, other (non-primary) home, and real-estate funds. The policy YAML carries one regime code per asset class:

$$\text{regime code}_k = \begin{cases} 0 & \text{step-up} \\ 1 & \text{carryover} \\ 2 & \text{deemed} \end{cases}$$

At each cell, the per-asset codes are resolved to a triplet $(\delta_{\text{vanish}}, \delta_{\text{route}}, \delta_{\text{realize}})$ by gain-stock-weighted averaging:

$$\delta_\bullet(a, t) = \frac{\sum_k G_{k,a,t}^B \mathbf{1}\{\text{code}_k = \bullet\}}{\sum_k G_{k,a,t}^B}, \quad \bullet \in \{\text{step-up, carryover, deemed}\}. \quad (6)$$

These shares feed the bathtub recurrence (route $\rightarrow$ heir routing, realize $\rightarrow$ deemed-at-death revenue, vanish $\rightarrow$ step-up). The holder-side $c_\phi(a, t)$ in the Bellman is $c_\phi = \theta \delta_{\text{route}} + 1 \cdot \delta_{\text{realize}} + 0 \cdot \delta_{\text{vanish}}$, with one additional adjustment on primary-home gains under deemed: see D.3 below.

D.3 §121 carve-out on primary-home gains

Under deemed realization, primary-home gains are taxed at death net of the §121 exclusion (\$250k single, \$500k joint). The per-record taxable primary-home gain is

$$g_i^{\text{pri},121} = \max\left(0, g_i^{\text{pri}} - \text{pref.kg_sec121_excl}_i\right), \quad (7)$$

and the cell-level adjustment to the deemed-channel base is the gain-stock-weighted utilization ratio $\rho^{121}(a, t) = G_{\text{pri},121}^B(a, t)/G_{\text{pri}}^B(a, t)$, applied within the regime mix on the primary-home asset class. §121 applies only to deemed; under carryover the basis carries to heirs with their own future eligibility, and under step-up the exclusion is moot.

D.4 Terminal-charity model

Bequests directed to qualifying charities are deductible from the estate and avoid the deemed-realization tax base. We model the share of decedent gain stock that is routed to charitable bequests as a two-stage logistic function of the decedent’s 2026-real estate value $E_{i,2026} = E_i \cdot \text{CPI-U}_{2026}/\text{CPI-U}_t$ (in millions):

$$p_i^{\text{ext}} = \sigma(\alpha_{\text{ext}} + \beta_{\text{ext}} \log E_{i,2026}), \quad (\alpha_{\text{ext}}, \beta_{\text{ext}}) = (-2.415, 0.458), \quad (8)$$

$$p_i^{\text{int}} = \sigma(\alpha_{\text{int}} + \beta_{\text{int}} \log E_{i,2026}), \quad (\alpha_{\text{int}}, \beta_{\text{int}}) = (-1.872, 0.468), \quad (9)$$

$$p_i^{\text{char}} = p_i^{\text{ext}} \cdot p_i^{\text{int}}, \quad (10)$$

where $\sigma(\cdot)$ is the logistic CDF, p^{ext} is the probability of any charitable bequest, and p^{int} scales the dollar share conditional on participation. Cell-level $\overline{p^{\text{char}}}(a, t)$ is dollar-of-gain-weighted; the deemed channel multiplies the cell’s decedent stock by $(1 - \overline{p^{\text{char}}})$ before computing taxable death revenue. Logistic coefficients are calibrated to IRS-SOI estate-tax data on the share of estates making charitable bequests and the share of those bequest dollars relative to gross estate.

D.5 SCF-derived heir distribution

Under carryover, decedent gain stock routes to heir cells through an age operator $\omega_{a \rightarrow h}$. Earlier specs used a Gaussian placeholder ($h \sim N(a - 30, 5^2)$); the production specification uses `resources/heir_distribution_scf2022.csv`, a dollar-weighted heir-age distribution constructed from the 2022 Survey of Consumer Finances inheritance records. The build script (`build_heir_distribution.R`) filters non-gift transfers, applies the Gale-Sabelhaus 2024 recency-of-receipt weights to convert the stock of past inheritances into a current-year flow, and dollar-weights heir ages. The resulting ω has heavier mass on ages 55–70 than the Gaussian placeholder, reflecting that high-wealth decedents are themselves older and route to relatively older heirs.

E Bathtub recurrence

E.1 State

The bathtub tracks $\Delta G_{a,t} = G_{a,t}^S - G_{a,t}^B$ on the cohort grid $a \in [18, 80]$. The recurrence (4) has two channels:

Survivor channel. $\sum_a A_{a \rightarrow h} (1 - m_{a,t}^{\text{eff}}) [(1 - r_{a,t}^S) \Delta G_{a,t} + G_{a,t}^B (r_{a,t}^B - r_{a,t}^S)]$. The bracket’s first term carries the previous-period ΔG forward net of survivor-realization; the second term is the new ΔG generated this period by the scenario realization-rate gap on the baseline stock. $A_{a \rightarrow h}$ is a one-year aging operator.

Heir-routing channel. $\delta_{\text{route}}(a, t) \sum_a \omega_{a \rightarrow h} m_{a,t}^{\text{eff}} (G_{a,t}^B + \Delta G_{a,t})$. Active only on the carryover share; routes the full decedent stock (baseline + deviation) through the SCF heir distribution to heir cells, where it joins the survivor channel in the subsequent period.

The effective cell mortality $m_{a,t}^{\text{eff}}$ in the bathtub is the same gain-stock-weighted quantity used in the Bellman.

E.2 Lock-in clamp

The cell-level revenue accounting includes a term $\tau^S r^S \Delta G$ that captures revenue on the policy-induced stock. Under permanent rate hikes ΔG can run sufficiently negative that $r^S \Delta G$ would subtract more from the realization base than the cell holds. The implementation clamps

$$\Delta G_{a,t}^{\text{eff}} = \max(\Delta G_{a,t}, -G_{a,t}^B), \quad (11)$$

so cell-level drawdowns cannot exceed full depletion of the baseline gain stock. The deemed-channel multiplier $\text{deemed_factor}(a, t) = \max(0, (G^B + \Delta G)/G^B)$ has the same non-negativity guarantee. The clamp matters only in extreme persistent-shock cohorts; in production scenarios it binds rarely.

E.3 Why an exogenous inflow can be omitted

Baseline gain-stock evolution carries an additive exogenous inflow (appreciation, new saving, portfolio-composition changes, imputation residual). For the baseline-to-scenario delta the bathtub actually tracks, that inflow cancels: if $G_{t+1}^j = H^j(G_t^j, r_t^j) + I_{t+1}$ for $j \in \{B, S\}$ with a common exogenous I_{t+1} , then $\Delta G_{t+1} = H^S(G_t^S, r_t^S) - H^B(G_t^B, r_t^B)$, and I_{t+1} drops out. So the bathtub never has to compute the inflow, and the Bellman's FOC is unaffected by it (it's not caused by today's marginal realization decision).

F Calibration procedure

F.1 Cell intercept recovery (inner κ inversion)

For each candidate ψ , the baseline Bellman runs backward over (a, t) . At each cell:

1. Compute continuation value $W_{a+1, t+1}^B$ (already solved).
2. Compute baseline marginal cost $MC_{a, t}^B = \tau^B + \beta(1 - m) W_{a+1, t+1}^B + \beta m F^B$.
3. Set $\kappa_{a, t} = MC_{a, t}^B + \psi r_{a, t}^{D, B}$, where $r^{D, B} = r^B - r_{\text{exog}, B}$. At corner cells with $r^{D, B} = 0$ set $\kappa = MC^B$.
4. Compute $W_{a, t}^B$ at the observed choice.

This is analogous to recovering cell fixed effects.

F.2 ψ bisection against the long-run target

For each candidate ψ and recovered $\{\kappa\}$, solve the scenario Bellman under a uniform +1pp perturbation, run the bathtub, and aggregate to a model semi-elasticity at simulation year 30:

$$\hat{\epsilon}^{\text{LR}}(\psi) = \frac{\log R_{30}^S - \log R_{30}^B}{0.01}. \quad (12)$$

Bisect ψ until $\hat{\epsilon}^{\text{LR}}(\psi) = -2.52$. The anchor at year 30 is deliberately past the bathtub's ~ 10 -year stock build-up so the calibration locks the long-run steady state rather than a transient.

F.3 ϕ_P bisection against the short-run target

For each candidate ϕ_P , re-solve the inner ψ bisection (so the long-run target stays satisfied), then run a delayed +5pp shock (announced at t , effective at $t + 1$) and compute the year-1 semi-elasticity:

$$\hat{\epsilon}^{\text{SR}}(\phi_P) = \frac{\log R_1^S - \log R_1^B}{0.05}. \quad (13)$$

Bisect ϕ_P until $\hat{\epsilon}^{\text{SR}}(\phi_P) = +5.04$. The outer bisection converges in ~ 10 – 20 iterations; warm-starting the inner ψ search from the previous outer iteration's solution keeps the total cost under an hour on a full-sample baseline.

F.4 Re-calibration triggers

Re-run the calibration whenever any of the following change: Tax-Data vintage, the discount factor series (β_t), bucket shares (ϕ_I, w_{ref} , timing window), the death-regime parameter θ , mortality input, the SCF heir distribution, or any Bellman primitive.

G Implementation notes

G.1 Two-pass solve

For each scenario, the bathtub pre-pass (`kg_dyn_run_bathtub_pass`) performs:

1. **Pass 1** (baseline) over the extended grid [18, 119]: recover $\kappa(a, t)$ and $W^B(a, t)$ from observed baseline realizations.
2. Per-year scenario regime mix (`kg_dyn_build_regime_mix`): resolve per-asset codes to $(\delta_{\text{vanish}}, \delta_{\text{route}}, \delta_{\text{realize}})$, compute c_ϕ , build the cell-level §121 adjustment.
3. **Pass 2** (scenario): solve the Bellman using fixed $\{\kappa\}$ and ψ to obtain $r^{D,S}(a, t)$.
4. Planned-timing schedule from $\tau^S - \tau^B$ (`kg_dyn_build_planned_timing`).
5. Per-year: combine buckets into $r_{a,t}^S$, run the bathtub step (`kg_dyn_step_recurrence`), build the cell-level state table, persist to disk.

The downstream behavior module (`config/scenarios/behavior/kg_dynamics/turnover.R`) is a pure allocator: for each tax-unit record in year t , it reads the year’s state table and translates cell quantities to per-record `kg_lt` adjustments via `kg_dyn_apply_to_records`.

G.2 Resources

- `resources/PerLifeTables_{M,F}_Alt2_TR2024.csv`: SSA Trustees Report Alt-2 life table for the 81–119 Bellman tail.
- `resources/heir_distribution_scf2022.csv`: SCF-2022 dollar-weighted heir-age distribution.

G.3 Out of scope

New TBL method does not currently model: (a) per-asset-class ψ heterogeneity, (b) age-varying ψ , (c) an embedded-gain-ratio cell-state dimension, (d) record-level longitudinal wealth histories, or (e) dynastic heir realization choices. Per-asset ψ is the most likely near-term extension; the asset-class infrastructure is already in place (`KG_DYN_ASSET_CLASSES`).